

EXP 2: For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in Python to output a description of the set of all hypotheses consistent with the training examples

AIM

To implement and demonstrate the Candidate Elimination Algorithm in Python using a training dataset stored in a CSV file to find the set of all hypotheses consistent with the given training examples.

SOFTWARE USED

- Google Colab
- Python 3.x
- Pandas Library

PROCEDURE

1. Create a CSV file containing the training dataset.
2. Upload the CSV file to Google Colab.
3. Read the CSV file using the Pandas library.
4. Initialise the Specific hypothesis (S) and General hypothesis (G).
5. Process each training example:
 - Update **S** for positive examples.
 - Update **G** for negative examples.
6. Display the final Specific and General hypotheses.

Step 1: Create the CSV File

Create a file named trainingdata.csv with the following content:

Sky, AirTemp, Humidity, Wind, Water, Forecast, EnjoySport

Sunny, Warm, Normal, Strong, Warm, Same, Yes

Sunny, Warm, High, Strong, Warm, Same, Yes

Rainy, Cold, High, Strong, Warm, Change, No

Sunny, Warm, High, Strong, Cool, Change, Yes

STEP 2: GOOGLE COLAB CODE

```
import pandas as pd
import numpy as np
# Read the dataset
data = pd.read_csv('trainingdata.csv')
# Separate attributes and target
concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])
# Candidate Elimination Algorithm
def candidate_elimination(concepts, target):
    specific_h = concepts[0].copy()
    general_h = [['?' for _ in range(len(specific_h))]
                  for _ in range(len(specific_h))]
    for i, h in enumerate(concepts):
        if target[i] == 'Yes':
            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    specific_h[x] = '?'
                    general_h[x][x] = '?'
        elif target[i] == 'No':
            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    general_h[x][x] = specific_h[x]
            else:
                general_h[x][x] = '?'
```

```
    general_h = [g for g in general_h if g != '?' * len(specific_h)]
    return specific_h, general_h
# Execute the algorithm
specific, general = candidate_elimination(concepts, target)
print("\nFinal Specific Hypothesis:")
print(specific)
print("\nFinal General Hypothesis:")
for g in general:
    print(g)
```

OUTPUT

```
Final Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' '?']

Final General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
```

RESULT

The Candidate Elimination Algorithm was successfully implemented in Google Colab using a CSV file, and the final Specific and General hypotheses consistent with the given training examples were obtained.